Lab 1 FDSAI

2025-2026

We will use the **NHANES (2015–2016)** dataset, which originates from the **National Center for Health Statistics (USA)**and is part of the **National Health and Nutrition Examination Survey** series of reports (https://wwwn.cdc.gov/nchs/nhanes/). The descriptions of the variables in the NHANES dataset can be found at the following link: https://wwwn.cdc.gov/nchs/nhanes/search/default.aspx.

The variables **BMXHT** and **BMXWT** that we will analyze below refer to the **height** of the surveyed subjects (in cm) and their **body weight** (in kg), respectively. We will also consider their **gender**, recorded in the binary variable **RIAGENDR** (coded as: 1 = male, 2 = female).

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1. Extract from the dataset only the information related to **gender**, **height**, and **body weight**.
   Check whether the new dataset contains any missing values (NaN) and remove them if necessary.
2. Plot **frequency histograms** for height and body weight — both **separately by gender** and for the **entire sample**.
3. Produce **descriptive statistics** (sample size, mean, variance, standard deviation, minimum, maximum, median, quartiles) for height and body weight — both **separately by gender** and for the **entire sample**. Comment on the results. Also, create a **scatter plot** for height versus body weight — both **separately by gender** and for the **entire sample**. Compute the **correlation coefficient** in each case. What do you observe?